CS 452/552: Operating Systems

Instructor

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Meetings

Lectures:	MoWe	10:30–11:45	CCP-221
Office hours:	MoWe	11:45–12:15	CCP-258
	TuTh	02:15–02:45	CCP-258
	or by appointment		CCP-258

Lectures will be audio/video recorded, and made available, afterwards.

I am happy to answer questions by email. Please please click here, or see:

> pub/doc/EmailQuestions.pdf

Catalog Description

Operating systems structure and design. Process management, concurrency and synchronization, interprocess communication, scheduling, device management, memory management, file systems and security. Case studies of multiple operating systems.

PREREQ: CS 230, CS 253, CS 321, ECE 330, and CS 155.

Goals

Students are introduced to basic concepts of operating systems, including:

- using processes and threads

- operating system organization
- computer organization
- device management
- implementing processes and threads
- scheduling
- synchronization
- interprocess communication
- deadlock
- memory management
- virtual memory
- file management
- security
- networks

Textbook

Operating Systems: Three Easy Pieces, by A. Arpaci-Dusseau and R. Arpaci-Dusseau, 2018, online.

Other Course Material

This syllabus, lecture slides, assignments, and other material is available in what we'll called our "`pub`" directory. It is available in three places: GitHub, Canvas, and the `onyx` cluster of computers. This directory is read-only. So, you might want to copy it, perhaps to your local computer.

The GitHub `pub` directory can be accessed at GitHub, as shown in:

    pub/GitHub

The Canvas `pub` directory can be accessed from our Canvas website, via the "Files" tab on the left sidebar.

The `onyx` `pub` directory can be accessed directly, by computers in our Computer Science Lab (CCP-240, CCP-241, and CCP-242).

Since `onyx.boisestate.edu` services Secure Shell (SSH) requests, you can also use SSH clients (e.g., `scp` and `sftp`) to access this `pub` directory remotely. However, beware: It contains symbolic links to parent directories, and `scp -r` will unconditionally follow them, thereby looping forever. To avoid this, use `sftp` or `tar`/`scp`, as needed.

In any case, the `onyx` `pub` directory is at:

```
~jbuffenb/classes/452/pub
```

The `onyx` cluster also has the advantage of containing all of the translators we will use.

Grades

At the end of the semester, the Registrar requires a letter grade for each student. Accordingly, during the semester, homework and exams are evaluated, and numeric scores are assigned. Each such artifact is worth a certain number of points, and has a weight. From these scores, a student's overall numeric/raw percentage is computed.

Collectively, a class's raw percentages form a distribution, with a mean and standard deviation, sometimes called a Bell Curve. An algorithm normalizes these values into a "grading" distribution, with a desired mean and standard deviation of 85 and 10 (i.e., the values are "curved"). A normalized value is then mapped to a letter grade, in the conventional way. If you are interested in the gory details, click here.

During the semester all of this evolving data can be found in our `pub` directory, on `onyx`, its nodes, Canvas, and GitHub. Homework is due at 11:59PM, Mountain Time, on the day it is due. Late work is not accepted. To submit your solution to an assignment, login to a lab computer, change to the directory containing the files you want to submit, and execute:

`submit jbuffenb` *class assignment*

For example:

```
submit jbuffenb cs452 hw1
```

The `submit` program has a nice `man` page.

When you submit a program, include: the source code, sample input data, and its corresponding results.

Scores are posted in our `pub/scores` directory, as they become available. You will receive a code, by email, indicating your row in the score sheet. You are encouraged to check your scores to ensure they are recorded properly. If you feel that a grading mistake has been made, contact me as soon as possible.

The weights of homework and exams is shown in the table below:

Activity	*Weight*
Homework	40%
Exam	30%
Final	30%

Homework

Several homework problems are assigned during the semester. Each asks you to develop software in the C programming language. Students work on these individually.

The rubric that will be used to grade each assignment is distributed with the assignment. Try to focus on the assignments, rather than the rubrics.

Exams

A midterm exam and a final exam are administered, near week eight, and at the end, of the semester. These are in-class, pencil-and-paper, open-notes, open-book (paper), no-laptop, and no-phone tests. Students work individually.

Graduate-Student Grades

This is a dual-listed section: containing undergraduate and graduate students.

Graduate students are assigned extra work: additional assignments, or additional parts of existing assignments. It may take the form of software development, or research and in-class presentation. Details are determined by the number of graduate students enrolled. In any event, rubrics will reflect this extra work.

For graduate students, the “grading” distribution, discussed above, is different. It has a mean and standard deviation of 80 and 10. The effect is that a graduate student is expected to “do better” than an undergraduate student.

Attendance

In-person lecture attendance is an important part of course participation. Attendance is taken at each lecture: starting five minutes before the scheduled start time, and ending fifteen minutes after the scheduled start time. Attendance is not taken during the first week of classes, holidays, or finals week.

Attendance can affect your grade. Each absence results in a one-percent reduction of your overall normalized percentage. Since a few absences are expected, completion of BSU's on-line end-of-semester course evaluation will erase up to five absences. Since evaluations are anonymous (at least to me), I will try to remind you to email me evidence of completing one: a screenshot will do.

Attendance is administered wirelessly, via the iClicker app, available for free, from your smartphone's app store. For more information, click here.

Source-Code Documentation

Good documentation and programming style is very important. Your programs must demonstrate these qualities for full credit. Good documentation and programming style includes:

- heading comments giving: author, date, class, and description
- function/procedure comments giving description of: purpose, parameters, and return value
- other comments where clarification of source code is needed
- proper and consistent indentation
- proper structure and modularity

For more information, and examples, click here.

Academic Integrity

The University's goal is to foster an intellectual atmosphere that produces educated, literate people. Because cheating and plagiarism are at odds with that goal, those actions shall not be tolerated in any form. Academic dishonesty includes assisting a student to cheat, plagiarize, or commit any act of academic dishonesty. Plagiarism occurs when a person tries to represent another person's work as his or her own or borrows directly from another person's work without proper documentation.

If a student engages in academic dishonesty, the student may be dismissed from the class and may receive a failing grade. Other penalties may include suspension or expulsion from the University.

For much more information about academic integrity, including examples of academic dishonesty, please click here. If you are unsure about a particular behavior, ask your instructor.

Generative Artificial Intelligence (AI)

You are allowed to use AI tools for studying course material, and for performing graded homework assignments, but not for taking exams.

If you use such a tool, do not let it replace your thinking and work. Ensure that you are fully engaging in learning and submitting original work. Cite any ideas, text, images, or other media generated by the tool. Include a brief description of how you used the tool.

Remember, you will not be able to rely on AI assistance during exams.

If you're unsure of whether or when to use AI tools in this course, ask your instructor.

Labs and Safety

Each student receives an account on the cluster of computers in the Computer Science Labs: CCP-240, CCP-241, and CCP-242. The cluster comprises a server named `onyx.boisestate.edu` and a set of nodes with shared home directories. It is remotely accessible, via SSH. The cluster runs the Linux operating system.

Physical access requires building and room access. After-hours building access, and all-hours room access, require an authenticated proximity-type student-identification card.

You are responsible for understanding and obeying lab rules.

Schedule

Week	*Date*	*Topic*	*Assigned*	*Due*	*Reading*
1	Aug 24 Mon	Introduction			1-2
	Aug 26 Wed	Processes and Processors	HW1		3-11
2	Aug 31 Mon				
	Sep 02 Wed				
3	Sep 07 Mon	Labor Day	HW2	HW1	
	Sep 09 Wed				
4	Sep 14 Mon				
	Sep 16 Wed				
5	Sep 21 Mon	Memory	HW3	HW2	12-24
	Sep 23 Wed				
6	Sep 28 Mon				
	Sep 30 Wed				
7	Oct 05 Mon	Concurrency			25-34
	Oct 07 Wed				
8	Oct 12 Mon				
	Oct 14 Wed			HW3	
9	Oct 19 Mon	Exam			
	Oct 21 Wed		HW4		
10	Oct 26 Mon				
	Oct 28 Wed				
11	Nov 02 Mon				
	Nov 04 Wed				
12	Nov 09 Mon	Input/Output Devices			35-42
	Nov 11 Wed		HW5	HW4	
13	Nov 16 Mon				
	Nov 18 Wed				
14	Nov 23 Mon	Thanksgiving			
	Nov 25 Wed	Thanksgiving			
15	Nov 30 Mon				
	Dec 02 Wed			HW5	
16	Dec 07 Mon				
	Dec 09 Wed				
17	Dec 16 Wed	Final: 9:30-11:30???			